Chapter 5
Project Analysis

5.1 Use Case Diagram

A use case diagram for AyurSpace highlights interactions between the Mobile User, Plant.id API, and Gemini API—authentication, plant scanning, dosha quiz, and AyurBot wellness advice.

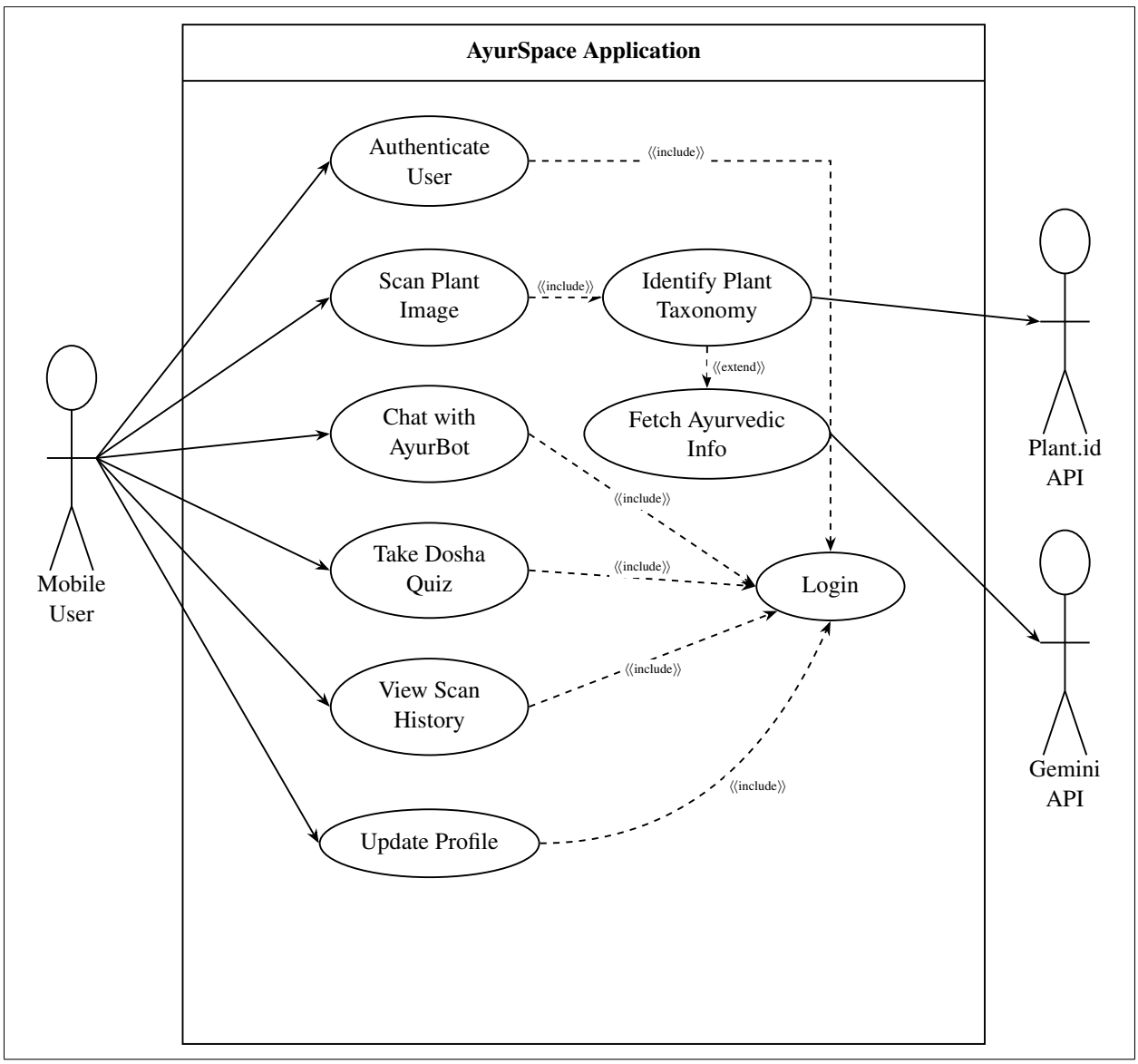

Figure 5.1: Use Case Diagram

5.1.1 Use Case Document

1. **User Registration / Authentication:**

   Firebase Authentication + Firestore profile sync for cross-device persistence.

2. **Scan Medicinal Plant:**

   Camera capture → local compression → Plant.id REST API → taxonomic identity and health assessment.

3. **Consult AyurBot:**

   Dynamic prompts inject user's *Dosha* profile + conversation history into Gemini LLM for hyper-personalized responses.

4. **Tridosha Questionnaire:**

   Algorithmic scoring of Vata/Pitta/Kapha questions generates user's *Prakriti*.

5. **Image Pre-processing (Internal):**

   Bicubic downsampling converts 12MP captures to compressed Base64 byte arrays, reducing API latency and ensuring stability.

5.2 Class Diagram

The class diagram adheres to UML 2.x conventions. Classes are colour-coded by stereotype: *service* (blue) for external-API wrappers, *entity* (green) for persistent domain objects, *value object* (purple) for immutable data carriers, and *state* (amber) for Riverpod notifier state. Visibility markers follow UML notation: (−) denotes private and (+) denotes public. Dashed arrows represent dependencies; solid arrows represent associations; a hollow diamond at the source denotes aggregation.

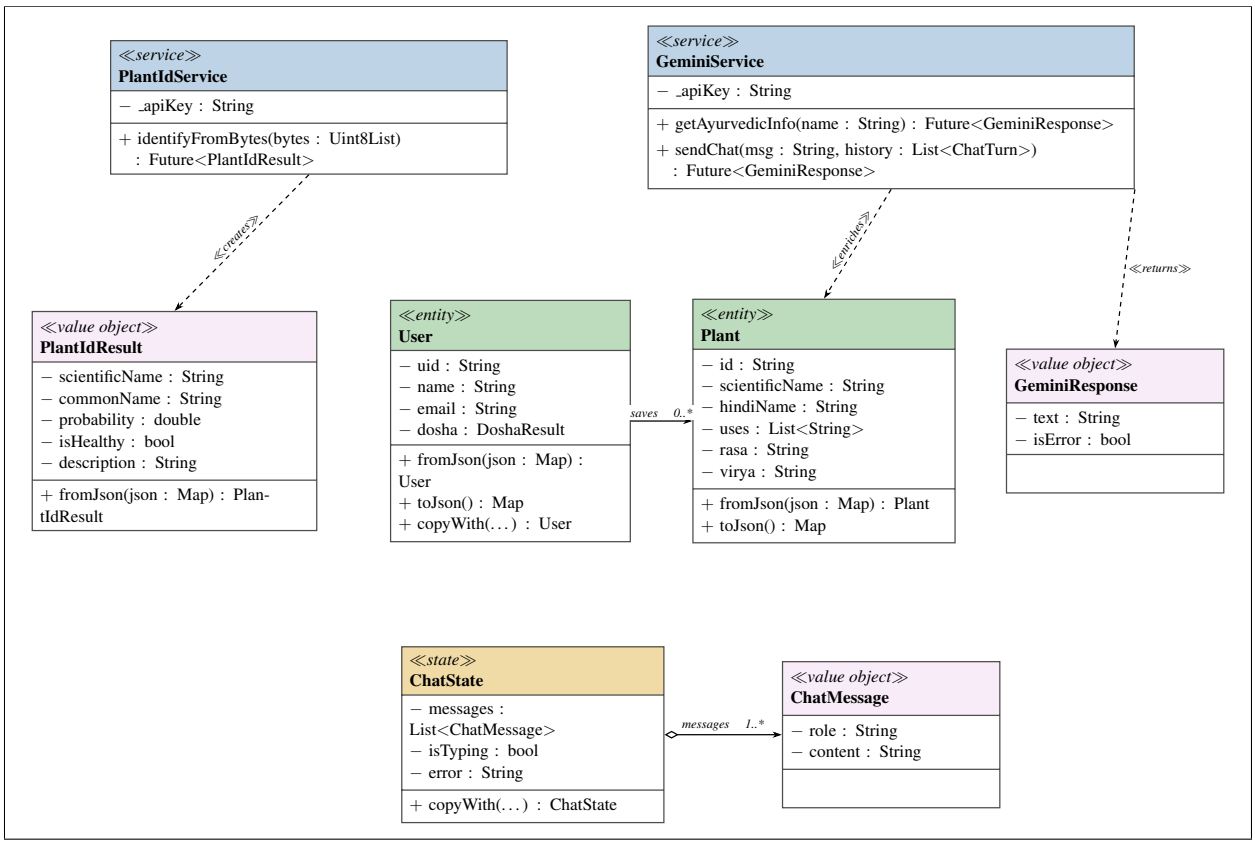

Figure 5.2: UML Class Diagram — AyurSpace System

5.3 Activity Diagram

The activity diagram captures the plant scanning flow: Dashboard $\rightarrow$ Camera $\rightarrow$ Compress $\rightarrow$ Plant.id API. If confidence < 0.2, error/retry. Otherwise $\rightarrow$ Gemini API $\rightarrow$ Firestore $\rightarrow$ UI render.

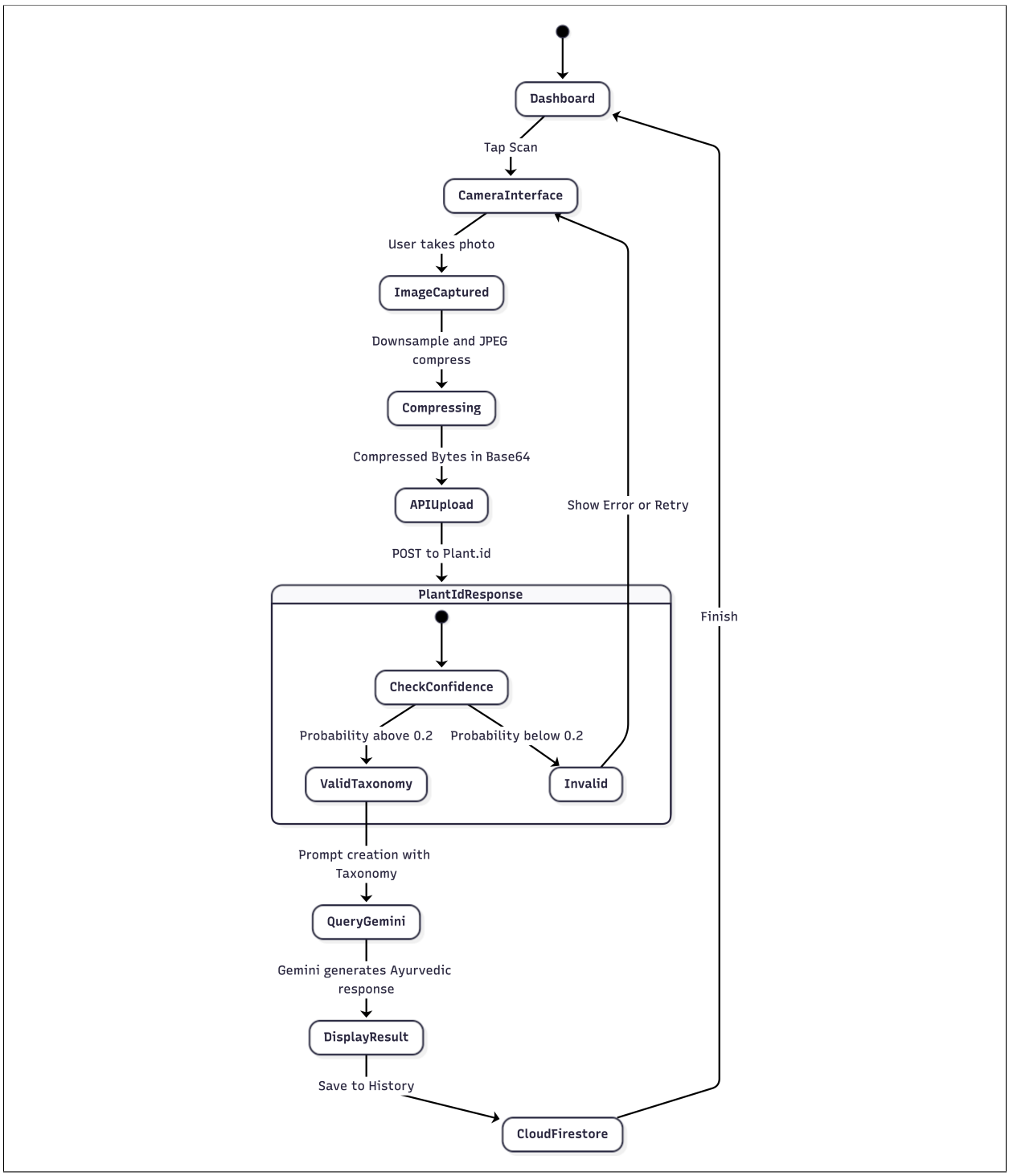

Figure 5.3: Activity Diagram

5.4 Sequence Diagram

The Sequence Diagram outlines the full execution timeline: Flutter Client → POST to Plant.id → taxonomy result → Gemini API prompt → JSON response → Firestore async write → UI dashboard update.

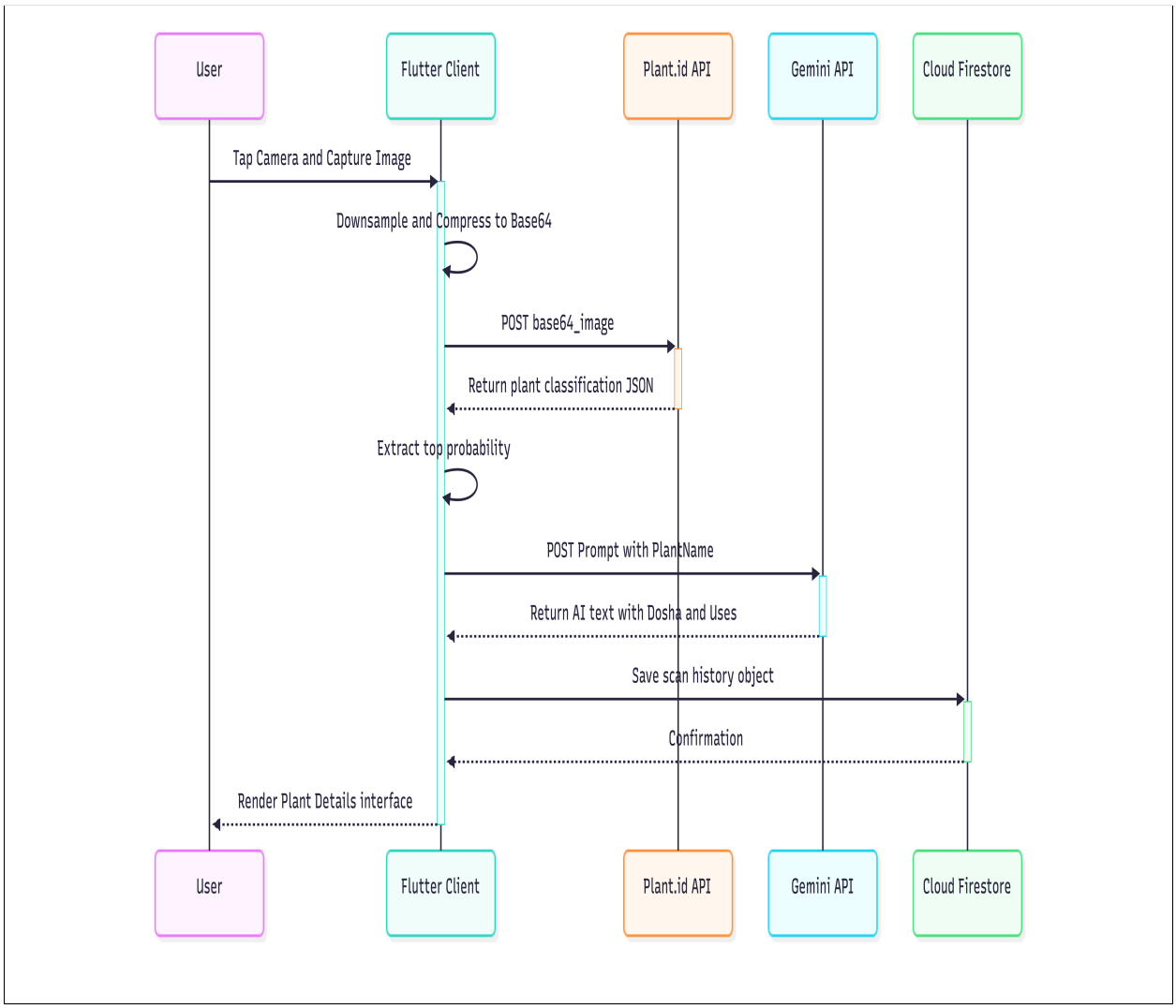

Figure 5.4: Sequence Diagram